

Fiscal-LPT Appendix Derivation Audit

Adversarial derivation check

May 13, 2026

1 Scope and verdict

This note audits the appendix derivations in `Fiscal-LPT/main.tex` against the live Julia implementation in `MCMS-private/src/build_equations.jl` and the current calibration block in `MCMS-private/src/calibration.jl`. The equations implemented in Julia are internally coherent under a precise set of conventions. The appendix mostly reports the implemented equations correctly, but several derivation statements are too loose. One item is a real derivation fault: the level government budget constraint written in the appendix is not the correct level ancestor of the implemented linear budget equation.

2 Adversarial findings

Fix: opening appendix conventions. The text says the new fiscal objects are log deviations. This is not true for $b_{k,t}^g$. The implemented debt variable is an additive real public-debt deviation normalized by steady-state GDP, not a log deviation. Treating debt as a log deviation would change the fiscal-rule and budget-constraint scaling.

Fix: labour-tax derivation and revenue equation. The appendix must define $\tau_{k,t}^N \equiv (1 - \widehat{T_{k,t}^N})^{-1}$, not just “the log labour wedge.” The wage Phillips curve and the labour-tax revenue equation both require this inverse after-tax wedge convention. If τ^N were instead $\widehat{1 - T^N}$, the signs would flip.

Fix: government budget derivation. The appendix starts from

$$B_t^g = (1 + i_{t-1}^g)B_{t-1}^g - P_t^C Y_t \text{Rev}_t.$$

That timing and gross-debt form do not linearize to the implemented equation

$$\beta b_t^g - \beta \bar{b}^g i_t^g + \text{Rev}_t = b_{t-1}^g - \bar{b}^g \pi_t^C.$$

The implemented equation comes from a one-period discount-debt budget,

$$Q_t^g B_t^g + P_t^C \bar{Y} \text{Rev}_t = B_{t-1}^g, \quad Q_t^g = (R_t^g)^{-1}.$$

This should be corrected in the appendix.

Clarify: revenue accounting. The revenue variables $rev^C, rev^N, rev^Y, rev^{tar}$ are not log revenues. They are first-order deviations of revenue scaled by steady-state GDP. The formulas are correct only under that additive normalization.

Clarify: consumption-tax Euler derivation. The Euler equation is correct, but the text should say that $i_{k,t}$ is the log deviation of the gross nominal rate, or the model’s first-order interest-rate

gap measured in the same units as inflation. Otherwise the coefficient on the interest-rate term is ambiguous.

Clarify: production-tax and subsidy experiments. A statement such as $\sigma_{4,21}^Y = -0.10$ is a shock-scale override. It is a ten percentage point subsidy only for a unit innovation and under the local mapping $dT^Y = (1 + \bar{T}^Y)\tau^Y$. If the model shock is interpreted as a statutory-rate movement, the gross-wedge conversion must be stated.

Verify: Dynare mirror claim. `main.tex` says there is a Dynare mirror in `b*_fiscal_lpt.mod`. The model equations in the mirror match the structure, but the parameter file still contains older placeholder assignments such as $\bar{T}^C = 0.20$, $\bar{T}^N = 0.30$, $\bar{b}^g = 0.60$, and $\psi^g = 0.01$. The current Julia calibration instead uses country-specific values and $\psi^g = 0$. Do not cite the Dynare parameter file as the current quantitative calibration unless it is regenerated or patched.

3 Notation that makes the derivations valid

For strictly positive variables, lower-case variables are log deviations:

$$x_t = \hat{X}_t = \log X_t - \log \bar{X}.$$

Debt is different. The implemented debt state is an additive deviation of real public debt from its steady-state ratio:

$$b_{k,t}^g = \frac{B_{k,t}^g}{P_{k,t}^C \bar{Y}_k} - \bar{b}_k^g.$$

It is normalized by current CPI and steady-state real GDP. This convention is forced by the implemented budget equation because no current $y_{k,t}$ term appears there.

The three tax variables are log deviations of gross wedges:

$$\tau_{k,t}^C \equiv \widehat{(1 + T_{k,t}^C)}, \quad (1)$$

$$\tau_{k,t}^N \equiv \widehat{(1 - T_{k,t}^N)}^{-1}, \quad (2)$$

$$\tau_{ki,t}^Y \equiv \widehat{(1 + T_{ki,t}^Y)}. \quad (3)$$

Therefore the first-order statutory-rate changes are

$$dT_{k,t}^C = (1 + \bar{T}_k^C)\tau_{k,t}^C, \quad (4)$$

$$dT_{k,t}^N = (1 - \bar{T}_k^N)\tau_{k,t}^N, \quad (5)$$

$$dT_{ki,t}^Y = (1 + \bar{T}_{ki}^Y)\tau_{ki,t}^Y. \quad (6)$$

These conversion formulas are the hidden step behind the revenue equations.

4 Consumption tax and the Euler equation

The household consumption FOC is

$$U_C(C_{k,t}) = \Lambda_{k,t} P_{k,t}^C (1 + T_{k,t}^C),$$

so, with CRRA utility $U_C(C) = C^{-\sigma}$,

$$\log \Lambda_{k,t} = -\sigma \log C_{k,t} - \log P_{k,t}^C - \log(1 + T_{k,t}^C).$$

After subtracting the steady state,

$$\lambda_{k,t} = -\sigma c_{k,t} - p_{k,t}^C - \tau_{k,t}^C.$$

The nominal bond FOC is

$$\Lambda_{k,t} = \beta R_{k,t} \mathbb{E}_t \Lambda_{k,t+1}, \quad R_{k,t} = 1 + i_{k,t}^{level}.$$

At first order, using $\beta \bar{R} = 1$ and ignoring Jensen terms,

$$\lambda_{k,t} = i_{k,t} + \mathbb{E}_t \lambda_{k,t+1},$$

where $i_{k,t}$ is the log deviation of the gross nominal rate. Substitute the multiplier expression:

$$-\sigma c_{k,t} - p_{k,t}^C - \tau_{k,t}^C = i_{k,t} + \mathbb{E}_t [-\sigma c_{k,t+1} - p_{k,t+1}^C - \tau_{k,t+1}^C].$$

Move terms and use $\pi_{k,t+1}^C = p_{k,t+1}^C - p_{k,t}^C$:

$$\sigma c_{k,t} = \sigma \mathbb{E}_t c_{k,t+1} - i_{k,t} + \mathbb{E}_t \pi_{k,t+1}^C + \mathbb{E}_t \tau_{k,t+1}^C - \tau_{k,t}^C.$$

Divide by σ :

$$c_{k,t} = \mathbb{E}_t c_{k,t+1} - \frac{1}{\sigma} (i_{k,t} - \mathbb{E}_t \pi_{k,t+1}^C - \mathbb{E}_t \tau_{k,t+1}^C + \tau_{k,t}^C).$$

This exactly matches the implemented Julia equation

$$c_k[0] = c_k[1] - \sigma^{-1} (i_k[0] - \pi_k^C[1] - \tau_k^C[1] + \tau_k^C[0]).$$

5 Labour tax and the wage Phillips curve

The static labour condition is

$$\frac{W_{k,t}}{P_{k,t}^C} = \frac{1 + T_{k,t}^C}{1 - T_{k,t}^N} C_{k,t}^\sigma N_{k,t}^\varphi.$$

Taking deviations gives

$$w_{k,t}^{flex} = \sigma c_{k,t} + \varphi n_{k,t} + \tau_{k,t}^C + \tau_{k,t}^N.$$

The sign on τ^N is positive because τ^N is the log deviation of the inverse after-tax labour wedge. A higher labour tax raises the pre-tax wage required by households.

The Calvo wage block uses the desired real-wage gap:

$$\pi_{k,t}^w = \kappa_k^w (w_{k,t}^{flex} - w_{k,t}) + \beta \mathbb{E}_t \pi_{k,t+1}^w.$$

Substitution yields

$$\pi_{k,t}^w = \kappa_k^w (\sigma c_{k,t} + \varphi n_{k,t} - w_{k,t} + \tau_{k,t}^C + \tau_{k,t}^N) + \beta \mathbb{E}_t \pi_{k,t+1}^w.$$

This matches the implemented wage equation. The only appendix problem is definition: without the inverse after-tax wedge convention, the sign is not well-defined.

6 Production tax and price setting

The selected instrument is a marginal-cost wedge:

$$MC_{ki,t}^{tax} = (1 + T_{ki,t}^Y) MC_{ki,t}.$$

Taking log deviations,

$$mc_{ki,t}^{tax} = mc_{ki,t} + \tau_{ki,t}^Y.$$

Hence every price Phillips-curve marginal-cost gap receives the same $+\tau_{ki,t}^Y$ term. For producer-currency pricing:

$$\pi_{lki,t} = \kappa_{ki}^p (mc_{ki,t} + \tau_{ki,t}^Y - p_{lki,t}) + \beta \mathbb{E}_t \pi_{lki,t+1}.$$

For local-currency pricing with $l \neq k$, the implemented relative-price term also includes the bilateral real exchange rate:

$$\pi_{lki,t} = \kappa_{ki}^p (mc_{ki,t} + \tau_{ki,t}^Y - p_{lki,t} - q_{kl,t}) + \beta \mathbb{E}_t \pi_{lki,t+1}.$$

For dominant-currency pricing, the q term is relative to the dominant currency country. The appendix is acceptable because it says it suppresses the currency-pricing indicators, but the full derivation should mention these branches if the appendix is meant to be self-contained.

7 Fiscal rules

For a generic wedge $x_{k,t}$, define a target

$$x_{k,t}^{target} = \phi_{y,k}^x y_{k,t} + \phi_{b,k}^x b_{k,t-1}^g.$$

Partial adjustment gives

$$x_{k,t} = \rho_k^x x_{k,t-1} + (1 - \rho_k^x) x_{k,t}^{target} + \sigma_k^x \varepsilon_{k,t}^x.$$

Therefore

$$\tau_{k,t}^C = \rho_k^C \tau_{k,t-1}^C + (1 - \rho_k^C) (\phi_{y,k}^C y_{k,t} + \phi_{b,k}^C b_{k,t-1}^g) + \sigma_k^C \varepsilon_{k,t}^C, \quad (7)$$

$$\tau_{k,t}^N = \rho_k^N \tau_{k,t-1}^N + (1 - \rho_k^N) (\phi_{y,k}^N y_{k,t} + \phi_{b,k}^N b_{k,t-1}^g) + \sigma_k^N \varepsilon_{k,t}^N, \quad (8)$$

$$\tau_{ki,t}^Y = \rho_{ki}^Y \tau_{ki,t-1}^Y + (1 - \rho_{ki}^Y) (\phi_{y,ki}^Y y_{k,t} + \phi_{b,ki}^Y b_{k,t-1}^g) + \sigma_{ki}^Y \varepsilon_{ki,t}^Y. \quad (9)$$

These equations match the Julia implementation. The only convention issue is again that b^g is an additive normalized debt deviation, not a log deviation.

8 Revenue linearizations

The implemented revenue variables are additive deviations normalized by steady-state GDP. They are not logs.

8.1 Consumption-tax revenue

Let

$$s_k^C = \frac{\bar{P}^C \bar{C}_k}{\bar{P}^C \bar{Y}_k} = \frac{\bar{C}_k}{\bar{Y}_k}.$$

Revenue is

$$\mathcal{R}_{k,t}^C = T_{k,t}^C P_{k,t}^C C_{k,t}.$$

Scaled by steady-state nominal GDP and abstracting from the common CPI level already handled in the real budget normalization,

$$rev_{k,t}^C = s_k^C \left[dT_{k,t}^C + \bar{T}^C c_{k,t} \right].$$

Using $dT^C = (1 + \bar{T}^C) \tau^C$,

$$rev_{k,t}^C = s_k^C \left[(1 + \bar{T}^C) \tau_{k,t}^C + \bar{T}^C c_{k,t} \right].$$

8.2 Labour-tax revenue

Let

$$s_k^L = \frac{\bar{W}_k \bar{N}_k}{\bar{P}^C_k \bar{Y}_k}.$$

Revenue is

$$\mathcal{R}_{k,t}^N = T_{k,t}^N W_{k,t} N_{k,t}.$$

The first-order scaled deviation is

$$rev_{k,t}^N = s_k^L \left[dT_{k,t}^N + T_{k,t}^N (w_{k,t} + n_{k,t}) \right].$$

Using $dT^N = (1 - T^N) \tau^N$,

$$rev_{k,t}^N = s_k^L \left[(1 - T_{k,t}^N) \tau_{k,t}^N + T_{k,t}^N (w_{k,t} + n_{k,t}) \right].$$

8.3 Production-tax revenue

Let

$$s_{ki}^{MCY} = \frac{\overline{MC}_{ki} \bar{Y}_{ki}}{\bar{P}^C_k \bar{Y}_k}.$$

Revenue is

$$\mathcal{R}_{k,t}^Y = \sum_i T_{ki,t}^Y MC_{ki,t} Y_{ki,t}.$$

The first-order scaled deviation is

$$rev_{k,t}^Y = \sum_i s_{ki}^{MCY} \left[dT_{ki,t}^Y + T_{ki,t}^Y (mc_{ki,t} + y_{ki,t}) \right].$$

Using $dT^Y = (1 + T^Y) \tau^Y$,

$$rev_{k,t}^Y = \sum_i s_{ki}^{MCY} \left[(1 + T_{ki,t}^Y) \tau_{ki,t}^Y + T_{ki,t}^Y (mc_{ki,t} + y_{ki,t}) \right].$$

This matches the implemented equation. If $T^Y < 0$, the base-effect term has the sign of a subsidy: higher marginal-cost output makes the subsidy more expensive.

9 Government borrowing spread

The spread equation is directly linear:

$$i_{k,t}^g = i_{k,t} + \psi_k^g b_{k,t-1}^g.$$

The baseline calibration currently sets $\psi_k^g = 0$ in Julia. Therefore the spread channel is present as an equation but shut down quantitatively in the current Julia baseline.

10 Government budget constraint

The implemented budget is a discount-debt equation. Start from

$$Q_{k,t}^g B_{k,t}^g + P_{k,t}^C \bar{Y}_k (rev_{k,t}^C + rev_{k,t}^N + rev_{k,t}^Y + rev_{k,t}^{tar}) = B_{k,t-1}^g.$$

Divide by $P_{k,t}^C \bar{Y}_k$:

$$Q_{k,t}^g \frac{B_{k,t}^g}{P_{k,t}^C \bar{Y}_k} + rev_{k,t}^C + rev_{k,t}^N + rev_{k,t}^Y + rev_{k,t}^{tar} = \frac{B_{k,t-1}^g}{P_{k,t}^C \bar{Y}_k}.$$

Use

$$\frac{B_{k,t}^g}{P_{k,t}^C \bar{Y}_k} = \bar{b}_k^g + b_{k,t}^g, \quad \frac{B_{k,t-1}^g}{P_{k,t}^C \bar{Y}_k} = \frac{P_{k,t-1}^C}{P_{k,t}^C} \frac{B_{k,t-1}^g}{P_{k,t-1}^C \bar{Y}_k}.$$

At first order,

$$\frac{P_{k,t-1}^C}{P_{k,t}^C} (\bar{b}_k^g + b_{k,t-1}^g) = \bar{b}_k^g + b_{k,t-1}^g - \bar{b}_k^g \pi_{k,t}^C.$$

For discount debt, $Q_{k,t}^g = 1/R_{k,t}^g$. With $\bar{Q}^g = \beta$ and $i_{k,t}^g$ as the log deviation of the gross government rate,

$$Q_{k,t}^g (\bar{b}_k^g + b_{k,t}^g) = \beta \bar{b}_k^g + \beta b_{k,t}^g - \beta \bar{b}_k^g i_{k,t}^g.$$

Subtract the steady-state identity $\beta \bar{b}_k^g = \bar{b}_k^g - \bar{rev}_k$. The first-order budget is

$$\beta b_{k,t}^g - \beta \bar{b}_k^g i_{k,t}^g + rev_{k,t}^C + rev_{k,t}^N + rev_{k,t}^Y + rev_{k,t}^{tar} = b_{k,t-1}^g - \bar{b}_k^g \pi_{k,t}^C.$$

This is the implemented equation. The current appendix should use this discount-debt parent equation. Starting from a lagged gross-interest budget is not just a harmless alternative presentation; it changes the timing and the linear coefficients.

11 Minimum edits recommended for main.tex

The appendix can be made mathematically tight with these targeted edits:

1. Define τ^C, τ^N, τ^Y exactly as gross-wedge log deviations and state that b^g is an additive debt-ratio deviation.
2. Replace the government budget parent equation with $Q_t^g B_t^g + P_t^C \bar{Y} rev_t = B_{t-1}^g$.
3. Say explicitly that revenue variables are first-order GDP-normalized deviations, not log revenue variables.
4. Qualify the Dynare-mirror sentence: the equation mirror matches the fiscal structure, but the current quantitative calibration is the Julia `src/calibration.jl` block unless the Dynare parameter file is updated.
5. In the policy-experiment paragraph, distinguish a wedge shock scale from a statutory-rate change.